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= Assignment 0 =

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Work type: individual

Grading system: 100 points

Delivery: git

Delivery time: kr 23.59, 04.09.2026, 20 point less for each day of delay (rounded up: 1 min delay → 1 day)

Task 0: (requirement)

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Make your own private git and share it with the lecturer.

Please invite me to your own repository (account user EnricoRiccardi)

Task 1: (10 pts (one file missing or different file/folder naming = 0 points)

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Make one folder: project_1_a

in the folder, construct 4 files:

File 1: code.py

File 2: test_code.py

File 3: tools.py

File 4: jupyther_code.ipynb

Task 2: (50 pts)
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In tools.py (15 pt)
----
write 2 functions:
one that takes as input a list of a list (e.g. [[], []., []])
one that takes as input a dict of dicts (e.g.{dict1: {}, dict2:{}}
and prints to screen in an descriptive way what there is inside the list and the dictionary (e.g. the output can look like: list 1 contains elements [1, 3, ‘mn’]
print to screen the total sum of the numerical values contained in the two object ([1, 3, ‘nm’] → 4)

note_1: this file should NOT contain the input lit and dicts.
note_2: functions without docstring = 0 points.

In code.py (15 pt)
----
import the functions from the tools.py file
define 2 objects: a list of list and a dict of dict
(feel free to use the fridge dictionary we discussed or to make up a new one).
Print to screen the content of these two objects by using the functions you constructed in tools.py (no copy and paste!)

In test_code.py (15 pt)
----
import the functions from the tools.py file
make 2 simple objects as in code.py
test your function outcome (function imported from tools) with assert

note: no need to test the print on screen functionality. Yet, how can we still test the code then?

In jupyther_code.`ipynb` (5 pt)

Redo what you did with the different python files in at least 3 different cells in a jupyter notebook (just copy and paste, it is for you to get used to this platform too).

Task 3 (40 pts)

Make a fridge management program

Make one folder: project_1_b with the same structure as task 1 and task 2

In tools.py write: (10 pts)

- a fridge management function:
Input:
temperature
open/close handle

*state your assumptions in the docstring!

Output:
power usage in 4 levels (0 to 3).

- a total power usage function
input: list (instantaneous values of power usage)
output: number (total power used)

In code.py (10 pts)
----
write the input
call the fridge management function
plot the instantaneous and cumulative power usage versus time
and the total power used (on screen).

In `test_code`.py (10 pts)
----
test each function

The jupyter notebook file is optional here.

Grading
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A: 90-100
B: 80-90
C: 70-80
D: 60-70
E: 50-60
F: 0-50

AI use:
======
Please don’t use it at this stage. The assignment can be solved with vanilla python.
The only external library needed is for plotting.

Bonus points:
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pylint score = 10, 5 pts
test with pytest = 5pts
submit 1 day earlier = 5 pts
use classes = 5 pts
use numpy to generate the data = 5 pts

note: bonus points do NOT count in the final averages